Light

GCSE Computer Science – Cyber Security Threats (Set 1)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Which statement best describes cyber security?

- ○ **A.** Software that speeds up internet connections
- ○ **B.** A type of computer virus that protects files
- ○ **C.** The processes, practices and technologies designed to protect networks, computers, programs and data from attack, damage or unauthorised access
- ○ **D.** A method for backing up files to the cloud

Q2 (1 mark)

What is social engineering in the context of cyber security?

- ○ **A.** Building social media websites
- ○ **B.** The art of manipulating people so they give up confidential information
- ○ **C.** Installing software on social networks
- ○ **D.** Designing computer networks for social clubs

Q3 (1 mark)

What does the term 'malware' mean?

- ○ **A.** An umbrella term for various forms of hostile or intrusive software
- ○ **B.** A specific type of computer virus only
- ○ **C.** Software that runs slowly on old computers
- ○ **D.** Programs designed to improve computer performance

Q4 (1 mark)

Which statement best describes phishing?

- ○ **A.** A technique for improving email delivery speeds
- ○ **B.** A method of catching fish using electronic devices
- ○ **C.** Software that blocks spam emails automatically

○ **D.** A technique of fraudulently obtaining private information, often using email or SMS

Q5 (1 mark)

What is blagging (also known as pretexting)?

○ **A.** A type of computer virus that displays advertisements

○ **B.** The act of creating and using an invented scenario to trick a victim into divulging information

○ **C.** A method of encrypting sensitive files

○ **D.** Software used to test network speeds

Q6 (1 mark)

What is shouldering (also called shoulder surfing)?

○ **A.** A type of malware that spreads through USB devices

○ **B.** A technique for improving typing speed

○ **C.** Observing a person's private information over their shoulder

○ **D.** A method of compressing large files

Q7 (1 mark)

How does a computer virus typically spread?

- ○ **A.** It attaches itself to another program or file and spreads when that program is executed
- ○ **B.** It automatically downloads from websites without any user action
- ○ **C.** It spreads through radio waves between nearby computers
- ○ **D.** It only affects computers that are connected to printers

Q8 (1 mark)

What is the key characteristic of a trojan?

- ○ **A.** It always displays warning messages before running
- ○ **B.** It can only infect mobile phones
- ○ **C.** It requires an internet connection to work
- ○ **D.** It disguises itself as legitimate software with a useful purpose but has a hidden malicious function

Q9 (1 mark)

What is the primary function of spyware?

- ○ **A.** To speed up computer processing
- ○ **B.** To create backup copies of important files
- ○ **C.** To gather information about a person or organisation without their knowledge

○ **D.** To block unwanted advertisements

Q10 (1 mark)

What is pharming?

○ **A.** A cyber attack intended to redirect a website's traffic to a fake website

○ **B.** A method of growing plants using computer-controlled systems

○ **C.** Software that organises files into folders automatically

○ **D.** A technique for speeding up database queries

Q11 (1 mark)

What is penetration testing?

○ **A.** Testing how fast data can travel through network cables

○ **B.** The process of attempting to gain access to resources without knowledge of usernames, passwords and other normal means of access

○ **C.** Checking whether computer monitors display colours correctly

○ **D.** Measuring how much weight a computer desk can support

Q12 (1 mark)

Why are default passwords a security threat?

- ○ **A.** They use too much computer memory
- ○ **B.** They make computers run more slowly
- ○ **C.** They cause software to crash frequently
- ○ **D.** They are widely known and published, making systems easy for attackers to access

Q13 (2 marks)

Sarah receives an email appearing to be from her bank, asking her to click a link and verify her account details urgently because of 'suspicious activity'. The email address shows 'security@y0urbank.com' (with a zero instead of the letter 'o'). What type of attack is this?

- ○ **A.** Shouldering
- ○ **B.** Phishing
- ○ **C.** Pharming
- ○ **D.** A legitimate security alert

Q14 (2 marks)

What is the key difference between how a virus spreads compared to a trojan?

- ○ **A.** Viruses only affect Apple computers while trojans only affect Windows
- ○ **B.** There is no difference; the terms mean exactly the same thing
- ○ **C.** A virus attaches to and spreads through other programs, while a trojan relies on users being tricked into downloading and installing it

- ○ **D.** Trojans are always beneficial while viruses are always harmful

Q15 (2 marks)

A company wants to test its security against both external hackers and potential malicious employees. What two types of penetration testing should they use?

- ○ **A.** Testing with no knowledge of credentials (simulating external attack) and testing with knowledge of basic credentials (simulating a malicious insider)
- ○ **B.** Testing only during daytime and testing only during nighttime
- ○ **C.** Testing using only Windows computers and testing using only Mac computers
- ○ **D.** Testing with antivirus enabled and testing with antivirus disabled

Q16 (2 marks)

Why is running unpatched or outdated software a cyber security threat?

- ○ **A.** Old software always runs faster than new software
- ○ **B.** Updates only add unwanted new features
- ○ **C.** Software patches never improve security
- ○ **D.** Known security vulnerabilities in older versions remain unfixed, allowing attackers to exploit them using publicly available methods

Q17 (2 marks)

A new employee is accidentally given administrator access to all company files, including confidential financial records they do not need for their job. What security threat does this represent?

- ○ **A.** Misconfigured access rights, where users have more permissions than their role requires
- ○ **B.** A phishing attack targeting the new employee
- ○ **C.** A virus infection on the company network
- ○ **D.** Shouldering by other employees

Q18 (2 marks)

An employee finds a USB drive in the car park and plugs it into their work computer to see what is on it. Why could this be a security threat?

- ○ **A.** USB drives always slow down computers significantly
- ○ **B.** The USB drive will automatically erase all files on the computer
- ○ **C.** The USB drive might contain malware that executes when plugged in, or files designed to trick the user into running malicious software
- ○ **D.** USB drives cannot store any data

Q19 (2 marks)

Which of these describes characteristics of weak passwords that make them security threats?

- ○ **A.** Passwords that contain numbers and symbols are always weak
- ○ **B.** Short passwords, common words, personal information, and predictable patterns that are easy for attackers to guess or crack

- ○ **C.** Any password longer than 8 characters is automatically weak
- ○ **D.** Passwords created by the user rather than generated by the computer

Q20 (2 marks)

How do blagging and phishing differ in their approach to targeting victims?

- ○ **A.** Blagging uses emails while phishing uses phone calls only
- ○ **B.** Blagging typically targets specific individuals with personalised scenarios, while phishing often targets large groups with generic messages hoping some will respond
- ○ **C.** They are exactly the same technique with different names
- ○ **D.** Phishing only works on mobile phones while blagging only works on computers

Q21 (2 marks)

What types of information might spyware collect and send back to attackers?

- ○ **A.** Only information the user explicitly agrees to share
- ○ **B.** Only the computer's serial number
- ○ **C.** Just the weather in the user's location
- ○ **D.** Browsing history, login credentials, keystrokes, emails, and documents accessed by the user

Q22 (2 marks)

Which practices help protect against social engineering attacks?

- ○ **A.** Training staff to recognise manipulation techniques, verifying requests through independent channels, and questioning unusual or urgent demands
- ○ **B.** Installing larger hard drives
- ○ **C.** Using faster internet connections
- ○ **D.** Only using computers manufactured after 2020

Q23 (2 marks)

Which combination of measures helps protect against malware infections?

- ○ **A.** Never turning the computer off and disabling all software updates
- ○ **B.** Keeping software updated, using antivirus software, being cautious about downloads and email attachments, and avoiding suspicious websites
- ○ **C.** Only using the computer on Tuesdays and Thursdays
- ○ **D.** Removing all security software to improve computer speed

Q24 (3 marks)

A company employee receives an email offering free software that claims to optimise computer performance. The employee downloads and installs it. Shortly after, the company notices unusual network activity and discovers that login credentials are being transmitted to an

external server. Which cyber security threats have occurred in this scenario?

- ○ **A.** Only pharming has occurred
- ○ **B.** The employee fell victim to a phishing email, installed a trojan (disguised as legitimate software), which contained spyware functionality (collecting and transmitting credentials)
- ○ **C.** Only a weak password attack has occurred
- ○ **D.** Only shouldering has occurred

Q25 (3 marks)

A company wants to test whether their security controls would detect and stop a disgruntled employee from accessing confidential data they are not authorised to see. Which type of penetration testing would be most appropriate, and why?

- ○ **A.** Testing without any credentials, because this simulates external hackers
- ○ **B.** No penetration testing is appropriate; only antivirus software is needed
- ○ **C.** Testing using only automated scanning tools with no human involvement
- ○ **D.** Testing with basic employee-level credentials (insider simulation), because this accurately models a legitimate employee attempting to exceed their authorised access

Q26 (3 marks)

An attacker phones a company receptionist, claiming to be from the IT department. They say a critical security update needs to be installed immediately and ask the receptionist to download software from a link

they provide. After installation, the company later discovers that confidential business data has been transmitted to external servers without authorisation. Identify the sequence of threats used in this attack.

- ○ **A.** Only pharming was used throughout the attack
- ○ **B.** Only weak password exploitation was involved
- ○ **C.** Blagging was used to deceive the receptionist into downloading a trojan, which contained spyware that secretly collected and transmitted company data
- ○ **D.** Only shouldering was employed in this attack

Q27 (3 marks)

A small business has the following issues: their router still uses the manufacturer's default password; employees share one login account for the accounting system; and their operating system has not been updated in 18 months. Which security threats do these issues expose the business to?

- ○ **A.** Default passwords allow easy router compromise; shared accounts mean misconfigured access rights and no individual accountability; outdated systems contain known vulnerabilities that attackers can exploit with readily available tools
- ○ **B.** These are all good security practices that reduce risk
- ○ **C.** Only the default password is a problem; the others are fine
- ○ **D.** These issues only affect computer speed, not security

Q28 (3 marks)

A hospital's computer system is infected with spyware that has been collecting and transmitting patient records to attackers for three months

before being detected. Analyse the potential impacts of this security breach.

- ○ **A.** The only impact is that computers may run slightly slower
- ○ **B.** There are no significant impacts because patient data is not valuable
- ○ **C.** The breach has minimal consequences since spyware is harmless
- ○ **D.** Severe impacts include: patient privacy violation breaching data protection laws; potential identity theft affecting patients; damage to hospital reputation and patient trust; possible regulatory fines; and attackers may sell or misuse sensitive health information

Q29 (3 marks)

A security audit identifies these issues at a company: (1) Some employees use passwords like 'password123'; (2) The company website has a known vulnerability that was patched six months ago but the patch has not been applied; (3) No staff have received social engineering awareness training. If the company can only address one issue immediately, which should be prioritised and why?

- ○ **A.** Social engineering training should wait because it takes too long to implement
- ○ **B.** The unpatched website vulnerability should be prioritised because it is a known, exploitable weakness with a publicly available fix, making it the most immediate and easily addressed critical risk
- ○ **C.** Weak passwords are not actually a security problem
- ○ **D.** None of these issues require attention because they are all minor concerns

Q30 (3 marks)

An attacker researches a company's CEO on social media, discovers they are attending a conference, and sends a personalised email appearing to be from the conference organisers asking the CEO to update their registration details via an attached form. The form is actually malware that, once opened, installs software to monitor all the CEO's computer activity. Identify all the cyber security threats present in this scenario and explain the attack chain.

- ○ **A.** This attack uses blagging (researched, personalised scenario targeting the CEO), delivered via phishing email (fraudulent message with malicious attachment), which installs a trojan (malware disguised as a legitimate form), containing spyware functionality (monitoring all computer activity)
- ○ **B.** Only shouldering is involved because the attacker used social media
- ○ **C.** This is only a pharming attack because it involves redirecting the CEO
- ○ **D.** No threats are present because the email appeared legitimate

Total: **55** marks • Time: **30 min**

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